RAFFLES INSTITUTION
H1 Mathematics 8864
2013 Year 6 Preliminary Examination

Time Allowed: 3 hours **Total Marks: 95**

Section A: Pure Mathematics [35 marks]

1 Find, algebraically, the set of values of k for which

$$kx^2 + 2x + k < 0$$

for all real values of x. [4]

Solution

$$\begin{aligned} kx^2 + 2x + k < 0 &\Rightarrow \quad k < 0 \quad \text{and} \quad 4 - 4k^2 < 0 \\ &\Rightarrow \quad k < 0 \quad \text{and} \quad (1-k)(1+k) < 0 \\ &\Rightarrow \quad k < 0 \quad \text{and} \quad k < -1 \text{ or } k > 1 \\ &\Rightarrow \quad k < -1 \end{aligned}$$

$\therefore\ \{k \in \mathbb{R}: k < -1\}$

2 Use a non-calculator method to solve the simultaneous equations:

$$y = \left(2^x + 7\right)\left(2^x - 7\right),$$
$$x = 1 + \log_2(y - 11).$$

[5]

Solution

$x = 1 + \log_2(y - 11) \Rightarrow 2^x = 2(y - 11)$

$\therefore\ y = (2y - 22 + 7)(2y - 22 - 7)$

$\Rightarrow 4y^2 - 89y + 435 = 0$

$\Rightarrow (y - 15)(4y - 29) = 0$

$\Rightarrow y = 15$ or $y = \dfrac{29}{4}$ (rejected, $y > 11$)

$y = 15$, $x = 1 + \log_2 4 = 3$

3 Integrate

(a) $\dfrac{2}{6x-7}$, [2]

(b) $3\left(\sqrt{x}+\dfrac{1}{\sqrt{x}}\right)^2$. [3]

Find the numerical value of the gradient of the curve $y = x\ln(2x+1)$ at the point where $x = 2$. Give your answer correct to 3 decimal places. [2]

Solution

(a) $\displaystyle\int \frac{2}{6x-7}\,\mathrm{d}x = \frac{1}{3}\ln\left|6x-7\right| + C$

(b) $\displaystyle\int 3\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2\mathrm{d}x = \int 3\left(x+2+\frac{1}{x}\right)\mathrm{d}x$

$$= \frac{3x^2}{2} + 6x + 3\ln\left|x\right| + C$$

From GC, gradient = 2.409 (3dp)

4 Consider the graph of $y = \dfrac{2-x}{x-k}$, where $k > 2$ is a positive constant.

Write down, in terms of k,

(i) the equation of the vertical asymptote, [1]

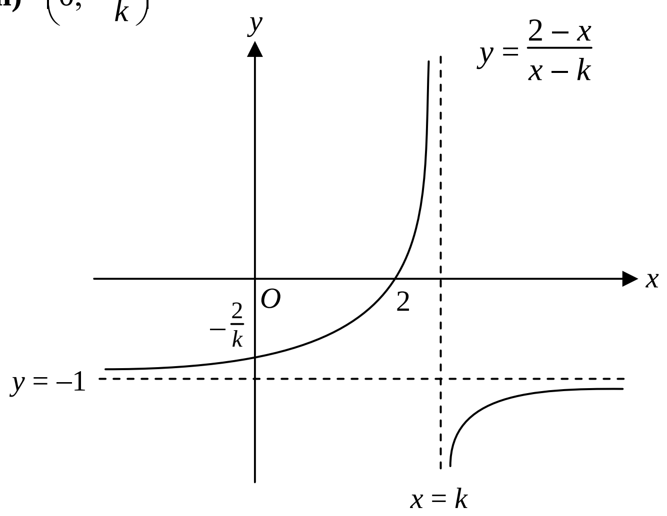
(ii) the coordinates of the intersection with the y-axis. [1]

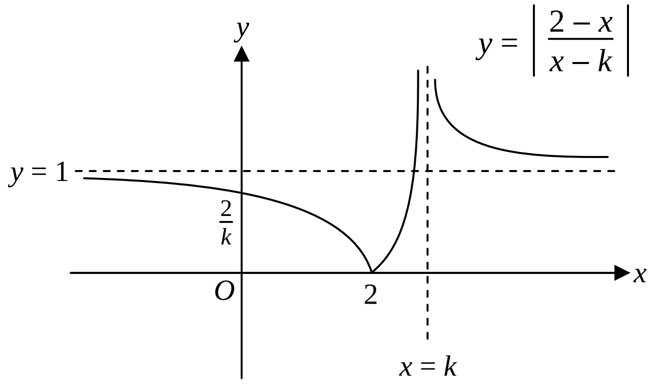
Sketch, on **separate diagrams**, the graphs of $y = \dfrac{2-x}{x-k}$ and $y = \left|\dfrac{2-x}{x-k}\right|$. You should indicate clearly on your graphs any points of intersection with the axes, turning points and asymptotes. [5]

Solution

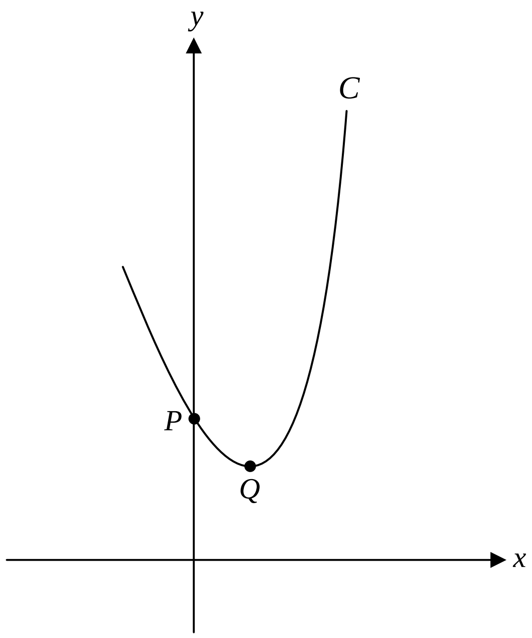
(i) $x = k$

(ii) $\left(0, -\dfrac{2}{k}\right)$





5 The curve C has equation $y = ae^{2x-1} - 4x + b$, where a and b are constants. The graph of C cuts the y-axis at P and has a minimum point at Q (see diagram below).



(i) By considering $\frac{d^2y}{dx^2}$, explain why a must be greater than zero. [2]

(ii) The tangent at P and the normal at Q intersect at the point $\left(\frac{1}{2}, \frac{4}{e} - 1\right)$. Show that $a = 2$ and find the value of b. [6]

(iii) Find the exact area of the region bounded by the curve C, the y-axis and the tangent at Q. [4]

Solution

(i) $y = ae^{2x-1} - 4x + b \Rightarrow \dfrac{dy}{dx} = 2ae^{2x-1} - 4$

$\Rightarrow \dfrac{d^2y}{dx^2} = 4ae^{2x-1}$

The graph of C has a minimum point at Q :

$\therefore \dfrac{d^2y}{dx^2} > 0 \Rightarrow 4ae^{2x-1} > 0$

$\Rightarrow a > 0$ (since $e^{2x-1} > 0$ for all real values of x)

(ii) $\dfrac{dy}{dx} = 0 \Rightarrow 2ae^{2x-1} - 4 = 0$

$x = \dfrac{1}{2}$ (normal at Q) $\Rightarrow 2ae^0 - 4 = 0 \Rightarrow a = 2$ (shown)

At point P,

$x = 0$, $y = 2e^{-1} + b$, $\dfrac{dy}{dx} = 4e^{-1} - 4$

$\therefore$ Equation of tangent is $y - (2e^{-1} + b) = (4e^{-1} - 4)(x - 0)$

At $\left(\dfrac{1}{2}, \dfrac{4}{e} - 1\right)$,

$(4e^{-1} - 1) - (2e^{-1} + b) = 2e^{-1} - 2 \Rightarrow b = 1$

(iii) Note that the coordinates of Q is $\left(\dfrac{1}{2}, b\right)$

$$\text{Area} = \int_0^{\frac{1}{2}} \{ (2e^{2x-1} - 4x + b) - (b) \} \, dx$$

$$= \left[e^{2x-1} - 2x^2\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} - e^{-1}$$

Section B: Statistics [60 marks]

6 In a certain school, each student is to take up exactly one of the following three courses: Baking (B102), Knitting (K106) or Reading (R107). The numbers of male and female students in each course are given in the following table:

	B102	K106	R107
Male	40	20	220
Female	160	80	80

One of the students is chosen at random. Find, giving your answers as a fraction in its lowest term, the probability that

(i) the student chosen is in B102, [1]

(ii) the student chosen is in K106, given that the student chosen is a female, [2]

(iii) the student chosen is a male, given that the student chosen is not in R107. [2]

Solution

(i) Required probability = $\frac{200}{600} = \frac{1}{3}$

(ii) Required probability = $\frac{80}{320} = \frac{1}{4}$

(iii) Required probability = $\frac{60}{300} = \frac{1}{5}$

7 The Environment ministry decides that a pilot study of a new initiative to tackle possible haze problem in schools will be undertaken before implementing the initiative nationwide. A sample of schools is to be selected to undergo this pilot study. To obtain this sample, a list of all 288 schools in the country is obtained and the schools are numbered alphabetically from 1 to 288. A number from 1 to 26 is then chosen using a random process. The corresponding school, and every 26th school thereafter, is included in the sample. For example, if the number 7 is chosen, then the schools numbered 7, 33, 59, … , n would be included in the sample.

(i) In the example given above, find the value of n, the number corresponding to the last school included in the sample. [1]

(ii) State the name given to this method of sampling. [1]

(iii) An advantage for this method of sampling is that it is unbiased. Explain why it is unbiased in this context. [1]

(iv) Give another advantage for this method of sampling in this context. [1]

(v) Comment on the size of the sample which would be obtained. [2]

Solution

(i) $n = 267$

(ii) Systematic sampling

(iii) Every school is equally likely (probability $= \frac{1}{26}$) to be selected.

(iv) The schools selected will be more evenly spread out in the country.

(v) Sample size can be 10 or 11, depending on the 1st number selected:
1 to 2 $\rightarrow$ sample size is 11, 3 to 26 $\rightarrow$ sample size is 10.

8 Events A and B are such that $P(A) = \frac{3}{7}$, $P(A \cap B) = \frac{1}{7}$ and $P(A' \cap B') = \frac{8}{21}$.

Find, in either order,

(i) $P(B)$,

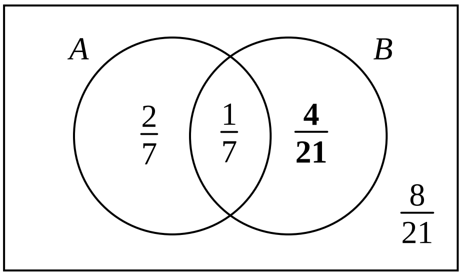
(ii) $P(A | B')$,

leaving your answers as a fraction in its lowest term. [5]

State, giving a reason, whether events A and B are independent. [2]

Solution

From information given:



Hence

(i) $P(B) = \frac{1}{7} + \frac{4}{21} = \frac{1}{3}$

(ii) $P(A | B') = \dfrac{P(A \cap B')}{P(B')} = \dfrac{\frac{2}{7}}{\frac{2}{7} + \frac{8}{21}} = \dfrac{3}{7}$

Since $P(A) \times P(B) = \frac{3}{7} \times \frac{1}{3} = \frac{1}{7} = P(A \cap B)$, events A and B are independent.

9 The ages (x months) and the heights (y centimetres) of 10 girls are given in the table below:

x	24	100	116	120	126	144	164	178	182	192
y	88	126	141	146	145	148	155	154	156	154

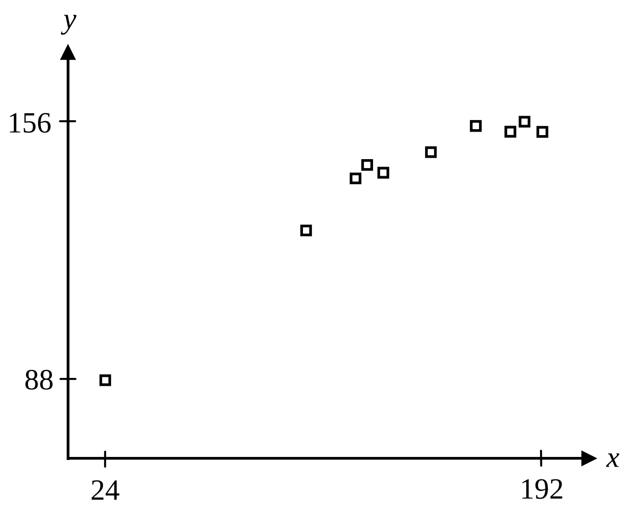
(i) Draw a scatter plot of the data. [2]

(ii) Calculate the linear correlation coefficient between x and y. [2]

(iii) Use an appropriate equation to estimate the height of a girl who is 26 years old. Comment on your result. [3]

Solution

(i)



(ii) From GC, $r = 0.940$ (3sf)

(iii) From GC, $y = 88.683 + 0.39091x$

When $x = 312$, $y = 210.65 = 211$ (3sf)

The estimate is not reliable as $x = 312$ is beyond the range of the observed data.

10 An experiment consists of throwing an ordinary fair die twice.

(i) Find the probability that the number '6' occurs at least once. [2]

(ii) By drawing a table of outcome, or otherwise, show that the probability that the sum of the scores on both throws exceeds 6 is $\frac{7}{12}$. [2]

Calculate the probability that, in 6 such experiments, there are more than 4 experiments in which the sum of the scores on both throws exceeds 6. [2]

Using a suitable approximation, calculate the probability that, in 60 such experiments, there are more than 40 experiments in which the sum of the scores on both throws exceeds 6. [4]

Solution

(i) Required probability $= 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$

(ii)

6	7	8	9	10	11	12
5	6	7	8	9	10	11
4	5	6	7	8	9	10
3	4	5	6	7	8	9
2	3	4	5	6	7	8
1	2	3	4	5	6	7
	1	2	3	4	5	6

Required probability $= \frac{21}{36} = \frac{7}{12}$ (shown)

Let X denote the number of experiments (out of 6) in which the sum of the scores on both throws exceeds 6. $X \sim \mathrm{B}\left(6, \frac{7}{12}\right)$

$\therefore$ $\mathrm{P}(X > 4) = 1 - \mathrm{P}(X \le 4) = 0.208$ (3sf)

Let Y denote the number of experiments (out of 60) in which the sum of the scores on both throws exceeds 6. $Y \sim \mathrm{B}\left(60, \frac{7}{12}\right)$

Since $n = 60$ is large, $np = 35 > 5$, $n(1-p) = 25 > 5$, $Y \sim \mathrm{N}\left(35, \frac{175}{12}\right)$ approx

$\therefore$ $\mathrm{P}(Y > 40) \overset{\text{cc}}{=} \mathrm{P}(Y > 40.5) = 0.0749$ (3sf)

11 An Automated Teller Machine (ATM) is located at the West end of a certain school. Each day, the number of users, x, is recorded and the data for a particular school term (10 weeks) is summarised as follow:

Number of users, x	0	1	2	3	4	5	6	7	8	9
Frequency, f	9	6	8	20	15	8	2	1	0	1

Show that the mean for this data is 3 and calculate its variance, giving your answer as a fraction in its lowest term. [3]

In an attempt to increase the number of users, the ATM is re-located to a central location in the school. The number of users per day is recorded for another school term (10 weeks) and the new data is summarised as follow:

$$\sum (x-3) = 28, \qquad \sum (x-3)^2 = 264.$$

Test, at the 4 % level of significance, whether the mean number of users per day has increased. State appropriate hypotheses for the test, and define any symbols used. [5]

Explain, in the context of the question, what is meant by "4 % level of significance". [2]

It is given instead that the new data recorded excludes all Saturdays and Sundays. Taking the population variance to be 7, find, to 1 decimal place, the minimum level of significance that would lead to a rejection of your null hypothesis. [2]

Solution

Mean = $\frac{\sum fx}{\sum f} = \frac{210}{70} = 3$ (shown)

From GC, Variance = $\frac{23}{7}$

Let μ denote the population mean.

Null hypothesis, $H_0 : \mu = 3$
Alternative hypothesis, $H_1 : \mu > 3$

Perform a <u>one-tail test</u> at the 4% level of significance.

Under H_0, $\overline{X} \sim N\left(\mu_0, \frac{s^2}{n}\right)$ approx by CLT,

where $\mu_0 = 3$, $s = 1.9141$, $n = 70$, $\overline{x} = 3.4$.

Using a z-test, p-value = 0.0402 (3sf).

Since p-value = 0.0402 > 0.04, we <u>do not reject</u> H_0 at the 4% level of significance.

Hence there is **insufficient** evidence at the 4% level of significance to conclude that the mean number of users per day has increased.

There is a probability of 0.04 that the test indicates that the mean number of users per day has increased when it is actually 3.

Assume X is normal, $\overline{X} \sim N\left(3, \frac{7}{50}\right)$

$\overline{x} = 3.56$, from GC, p-value = 0.067241

Hence the minimum level of significance that would lead to a rejection of the null hypothesis is 6.8%

12 The time for Mr. Teo to travel to work may be taken to have a normal distribution X with mean 14 minutes and standard deviation $\sqrt{6}$ minutes.

Find the probability that, for two consecutive working days, Mr. Teo's travel time

(i) is less than 13 minutes on each day, [2]

(ii) is less than 26 minutes altogether. [2]

Explain briefly why the answer to **(ii)** is greater than the answer to **(i)**. [1]

The time for Mr. Lee to travel to work may be taken to have an independent normal distribution Y with mean μ minutes and standard deviation σ minutes.

(iii) Given that $\mathrm{P}(Y > 13) = 0.00135$, find an equation involving both μ and σ. [2]

(iv) Given further that $\mathrm{P}\left(Y < \frac{1}{2}(X_1 + X_2)\right) = 0.97725$, where X_1 and X_2 are two independent observations of X, calculate the values of μ and σ, giving your answers to the nearest integer. [5]

(v) Give, in the context of the question, an interpretation of $\mathrm{P}\left(Y < \frac{1}{2}(X_1 + X_2)\right)$. [1]

Solution

Let X_1 and X_2 be the travel time for the two days.

(i) $X_1 \sim \mathrm{N}(14, 6)$, $X_2 \sim \mathrm{N}(14, 6)$

Required probability = $\mathrm{P}(X_1 < 13)\,\mathrm{P}(X_2 < 13) = 0.117$ (3sf)

(ii) $X_1 + X_2 \sim \mathrm{N}(28, 12)$

Required probability = $\mathrm{P}(X_1 + X_2 < 26) = 0.282$ (3sf)

The event described in **(i)** is a subset of the event described in **(ii)**.

$Y \sim \mathrm{N}(\mu, \sigma^2)$

(iii) $\mathrm{P}(Y > 13) = 0.00135 \Rightarrow \mathrm{P}\left(Z > \dfrac{13-\mu}{\sigma}\right) = 0.00135$

From GC, $\dfrac{13-\mu}{\sigma} = 3 \Rightarrow \mu + 3\sigma = 13$ -- (1)

(iv) $\frac{1}{2}(X_1 + X_2) \sim \mathrm{N}(14, 3)$, $Y - \frac{1}{2}(X_1 + X_2) \sim \mathrm{N}(\mu - 14, \sigma^2 + 3)$

$\mathrm{P}\left(Y < \frac{1}{2}(X_1 + X_2)\right) = \mathrm{P}\left(Y - \frac{1}{2}(X_1 + X_2) < 0\right) = 0.97725$

$\Rightarrow \mathrm{P}\left(Z < \dfrac{14-\mu}{\sqrt{\sigma^2 + 3}}\right) = 0.97725$

From GC, $\dfrac{14-\mu}{\sqrt{\sigma^2+3}} = 2$ -- (2)

Solving (1) and (2), $\mu = 10$, $\sigma = 1$ (nearest integers)

(v) It is the probability that the travel time for Mr. Lee on a particular day is less than the **average** travel time for Mr. Teo on two particular days.

— End of Paper —